

NAME: Diya Shah

Assignment: DHCP Server Configuration in Windows Server

1. Create a DHCP Scope Named 'CampusWiFi'

Steps:

1. Open Server Manager.
2. Go to Tools → DHCP.
3. Expand your server name.
4. Expand IPv4.
5. Right-click IPv4 and select New Scope.
6. Click Next.
7. Enter the scope name as CampusWiFi.
8. Configure the following:
 - Start IP Address: 192.168.50.100
 - End IP Address: 192.168.50.200
 - Subnet Mask: 255.255.255.0
9. Click Next.
10. Leave exclusions empty (unless required).
11. Continue with the default settings.
12. Activate the scope by selecting Yes, I want to activate this scope now.
13. Click Finish.

Result: A DHCP scope named **CampusWiFi** is successfully created with the IP address range **192.168.50.100 – 192.168.50.200**.

2. Create a DHCP Reservation

Steps:

1. In the DHCP Console, expand:
 - IPv4 → CampusWiFi → Reservations
2. Right-click Reservations and select New Reservation.
3. Enter the following details:
 - Reservation Name: Laptop

- IP Address: 192.168.50.150
 - MAC Address: *(Enter your laptop's MAC address, e.g., 00-15-5D-01-23-45)*
 - Description: Reserved IP for Laptop
4. Select Both for supported types.
 5. Click Add and then Close.

Result: The specified device will always receive the IP address 192.168.50.150 from the CampusWiFi DHCP scope.

3. Change the DHCP Lease Duration to 2 Days

Steps:

1. Right-click the CampusWiFi scope.
2. Select Properties.
3. Open the General tab.
4. Under Lease Duration for DHCP Clients, change the value to:
 - 2 Days
 - 0 Hours
 - 0 Minutes
5. Click Apply and then OK.

Result: The DHCP lease duration is successfully changed from the default value to 2 days.

4. Verify the Client Receives an IP Address

Steps:

1. Connect the client computer or virtual machine to the same network.
2. Open Command Prompt.
3. Run the following commands:

```
ipconfig /release
ipconfig /renew
ipconfig
```
4. Verify that the client receives an IP address within the range:
 - 192.168.50.100 – 192.168.50.200
5. If using the reserved device, verify it receives:
 - 192.168.50.150

6. Take a screenshot of the **ipconfig** output or the **Network Settings** showing the assigned IP address.

Result: The client successfully receives an IP address from the **CampusWiFi** DHCP scope.

Screenshot: *(Insert screenshot showing the assigned IP address.)*

5. Documentation of the DHCP Setup

Include the following in your **Word or PDF report**:

Title

DHCP Server Configuration – CampusWiFi

Contents

1. Installation of the DHCP Server role.
2. Creation of the **CampusWiFi** scope.
3. Configuration of the IP address range:
 - **192.168.50.100 – 192.168.50.200**
4. Subnet Mask:
 - **255.255.255.0**
5. DHCP Reservation:
 - Reserved IP: **192.168.50.150**
 - Device MAC Address: *(Your laptop's MAC address)*
6. Lease Duration changed to **2 Days**.
7. Verification of the client receiving an IP address.
8. Screenshots of:
 - DHCP Scope Creation
 - Reservation Configuration
 - Lease Duration Settings
 - Client IP Address (ipconfig or Network Settings)

Result: The DHCP configuration is fully documented with the required screenshots.

```
Administrator: C:\Windows\system32\cmd.exe
C:\Users\Administrator>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Default Gateway . . . . . : 

C:\Users\Administrator>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : server.com
    IPv4 Address. . . . . : 192.168.1.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : client1
    Primary Dns Suffix . . . . . : server.com
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : server.com

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : server.com
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-C0-E1-A3
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.1.50(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 29 June 2026 14:28:39
    Lease Expires . . . . . : 30 June 2026 14:28:38
```

File Action View Help			
DHCP			
server.server.co	Start IP Address	End IP Address	Description
IPv4	192.168.1.50	192.168.1.100	Address range for distribution
Server			
Scope			
Adc			
Adc			
Res			
Sc			
Pol			
Policies			
Filters			
IPv6			